

BloodChain AI — Simple Operational Guide & User Manual

> **System Note:** BloodChain AI is a development prototype and demonstration benchmark. All hospital names, inventory numbers, and scenarios are synthetic demonstration data.

1. What is BloodChain AI?

BloodChain AI is a regional blood supply coordination platform. It solves two critical problems: 1. **Emergency Shortages:** Hospitals running out of blood during trauma influxes. 2. **Preventable Wastage:** Blood expiring on hospital shelves because newer stock was used first.

It connects **hospitals**, **central blood banks**, **approving medical officers**, and **courier couriers** into one system to coordinate blood requests, clinical approvals, batch-level cold transport, and audit tracking.

2. Who Uses BloodChain AI? (The 5 Roles)

Every role has a strict job and cannot perform actions reserved for other roles:

Role & Persona	Where they work	What they can & cannot do
1. HOSPITAL STAFF (Dr. Rajesh Kumar)	Emergency Trauma Wards & Surgery Theaters	✔ Request emergency blood ✔ Record patient ward usage ✘ CANNOT approve transfers
2. AUTHORIZED APPROVER (Dr. Sarah Jenkins)	Regional Transfusion Medical Council	✔ Approve transfer requests ✔ Reject with mandatory reason ✘ CANNOT request blood
3. BLOOD BANK STAFF (Ananya Roy)	Central Blood Centers & Storage Hubs	✔ Inspect FEFO picklists ✔ Manage testing & batches ✘ CANNOT approve requests
4. LOGISTICS COURIER (Vikram Sethi)	Kaveri Cold-Chain Transport Fleet	✔ Start transport (dispatch) ✔ Confirm delivery receipt ✘ CANNOT approve transfers
5. ADMINISTRATOR (R. Administrator)	System Operations & Regional Audit	✔ Manage facility directory ✔ Review immutable audits

3. The Emergency Story: Step-by-Step

To see how BloodChain AI works in real life, follow this emergency highway trauma scenario:

The Problem

- At 02:15 AM, two highway accident victims arrive at **Manapparai Trauma Unit**.
- The surgeon needs **4 units of O-Negative blood** immediately.
- Manapparai has only 1 unit in stock, already reserved for an ICU patient (**0 usable units**).

Step 1: Hospital Requests Blood

- **Who:** Dr. Rajesh Kumar (`HOSPITAL\_STAFF`).
- **Where:** Screen: **Request Blood** (`/requests`).
- **Action:** Clicks + **Request Emergency Blood**, selects `O\_NEGATIVE`, quantity `4`, priority `EMERGENCY`.
- **Result:** Transfer created with status **Waiting for approval** (`PENDING\_APPROVAL`).

Step 2: System Checks "Safe-to-Share"

- **Who:** BloodChain AI platform (automatic).
- **Rule:** A donor hospital is **never** asked to give blood if doing so risks its own patients.
- **Calculation:**

$$Safe-to-Share = Available - Reserved - Safety\ Buffer$$
- **Result:** **Tiruchirappalli Central Hub** has 20 units total. After protecting its own safety buffer, it has **14 Safe-to-Share units**. The system recommends it as the donor.

Step 3: Medical Officer Approves

- **Who:** Dr. Sarah Jenkins (`AUTHORIZED\_APPROVER`).
- **Where:** Screen: **Approvals Queue** (`/command-center`).

- **Action:** Dr. Sarah reviews the trauma urgency and the donor hub's safety margin, then clicks **Approve Transfer**.
 - **Result:** Status changes to **Approved for dispatch** ('APPROVED'). An email alert is staged for dispatch.
(Alternative): *If Dr. Sarah clicks **Reject**, she **must** enter a clinical reason (e.g., "Alternative B+ blood available locally"*).* Leaving the reason blank is blocked.
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Step 4: Blood Bank Packs the Soonest-Expiring Batch (FEFO)

- **Who:** Ananya Roy ('BLOOD_BANK_STAFF').
 - **Where:** Screen: **Batches & Stock** ('/inventory').
 - **Rule: FEFO (First Expiry, First Out).** Always use blood that expires soonest.
 - **Action:** The system generates a picklist:
* `Batch A` (expires in 5 days): **Pick 2 units** (depleted first!). * `Batch B` (expires in 25 days): **Pick 2 units**.
 - **Result:** Zero wastage. The cooler box is packed and sealed for the courier.
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Step 5: Logistics Courier Dispatches

- **Who:** Vikram Sethi ('LOGISTICS_STAFF').
 - **Where:** Screen: **Cold Transport** ('/logistics').
 - **Action:** Vikram collects the box and clicks **Start Transport**.
 - **Result:**
* Status changes to **Being transported** ('IN_TRANSIT'). * Central Hub's available stock **immediately drops from 20 to 16 units**.
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Step 6: Destination Receives the Blood

- **Who:** Vikram Sethi ('LOGISTICS_STAFF') hands over to Dr. Rajesh.
 - **Where:** Screen: **Active Shipments** ('/requests').
 - **Action:** Clicks **Confirm Delivery**.
 - **Result:**
* Status changes to **Received by destination** ('RECEIVED'). * Manapparai Trauma Unit's stock **immediately increases from 1 to 5 units**. * The patient receives the transfusion in surgery.
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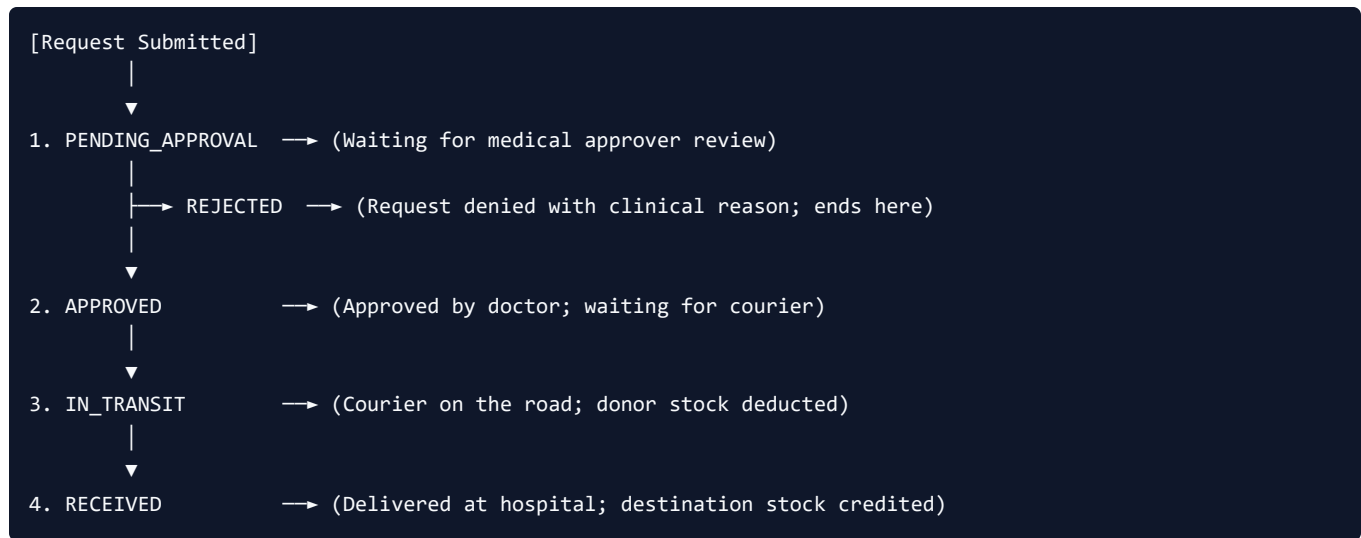
Step 7: Inventory Reconciled & Audited

- **Conservation Check:** Total blood units across both hospitals before transfer ($\$20 + 1 = 21\$$). Total units after transfer ($\$16 + 5 = 21\$$). **Zero phantom units created or lost.**
- **Audit History:** Screen: **Audit History** ('/audit-logs'). An unchangeable log records:

1. `TRANSFER\_CREATED` by Dr. Rajesh Kumar. 2. `TRANSFER\_APPROVED` by Dr. Sarah Jenkins. 3. `TRANSFER\_DISPATCHED` by Vikram Sethi. 4. `TRANSFER\_RECEIVED` by Vikram Sethi.

4. The 5 Transfer Statuses

BloodChain AI moves every request through a strict sequence:



- **Cannot skip steps:** A courier cannot dispatch blood before an approver signs off.
 - **Cannot double-credit:** Clicking "Confirm Delivery" twice will not add extra blood.
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5. Key Safety Rules

1. Safe-to-Share Guardrail

A hospital will **never** be stripped of blood needed for its own local emergencies. If a hospital needs 6 units for safety, those 6 units are locked and cannot be transferred away.

2. FEFO (First Expiry, First Out)

The algorithm automatically selects blood bags that will expire earliest so they are used first. Newer blood remains in storage.

3. Separation of Duties (No Self-Approval)

- A doctor requesting blood **cannot** approve their own request.

- A courier **cannot** change blood amounts or approve requests.
- An administrator **cannot** bypass doctors to approve transfers.

4. Zero Patient Privacy Risks (No PHI)

Email notifications and audit logs contain **only logistical information** (Hospital names, blood type, quantity, urgency). Patient names, bed numbers, and medical conditions are never sent in emails or exposed.

6. How the AI Prediction & Solver Work

- **Demand Forecasting:** Uses an `XGBoost` model trained on synthetic historical blood usage to forecast how many units each hospital will need over the next 24 hours.
- **Shortage Detection:** If projected usage exceeds current usable stock, the system raises an alert (`CRITICAL` or `WARNING`).
- **Route & Allocation Solver:**
* In the full design, Google OR-Tools calculates optimal multi-hospital routes. * In this local prototype, OR-Tools is uninstalled, so the system uses a built-in **Greedy Heuristic Fallback** (prioritizing lowest risk and shortest travel time).
Note*: While the API response shows `OPTIMAL`, mathematical global optimality is a future enhancement when OR-Tools is installed.

7. Email Notifications: "Submitted" vs "Delivered"

BloodChain AI uses Brevo Transactional Email to alert staff:

- **`submitted`**: Brevo accepted the email into its delivery queue. (This happens immediately).
 - **`delivered`**: An authenticated webhook confirms the email reached the inbox.
- Prototype Note*: In local testing without a public web address, emails are marked `submitted`. The system never pretends an email was delivered without proof.

8. Common Troubleshooting

Issue / Error	Why it happened	What to do
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Issue / Error	Why it happened	What to do
"Unauthorized Role" (403)	You tried to do an action not allowed for your role (e.g. Hospital Staff trying to approve).	Switch to the correct role using the top-right menu (e.g. Authorized Approver).
"Duplicate Transfer Request" (400)	A transfer for the same hospital and blood type is already active.	Wait for the active shipment to finish before requesting another.
"Reason Mandatory" (400)	You clicked Reject but left the explanation blank.	Type a clinical reason (e.g. " <i>Alternative stock available</i> ") and confirm.
"Exceeds Safe-Share" (400)	The donor hospital would be left with dangerously low stock.	Request a smaller quantity or pick another donor facility.

9. Quick Cheat-Sheet for New Users

Need to request blood?	→ Switch to HOSPITAL STAFF → Go to /requests → Click "+ Request Blood"
Need to approve a request?	→ Switch to AUTHORIZED APPROVER → Go to /command-center → Click "Approve"
Need to pick blood bags?	→ Switch to BLOOD BANK STAFF → Go to /inventory → Check FEFO Picklist
Need to start courier trip?	→ Switch to LOGISTICS STAFF → Go to /logistics → Click "Start Transport"
Need to confirm arrival?	→ Switch to LOGISTICS STAFF → Go to /requests → Click "Confirm Delivery"
Need to record surgery usage?	→ Switch to HOSPITAL STAFF → Go to /inventory → Click "Record Ward Usage"
Need to inspect audit trail?	→ Switch to ADMINISTRATOR → Go to /audit-logs

10. Prototype Limitations

1. **Synthetic Data:** All facilities, batches, and patient transfers are demo simulations. 2. **No EHR Integration:** BloodChain AI does not pull directly from live hospital record systems. 3. **Medical Supervision Required:** BloodChain AI does not perform physical laboratory cross-matching or replace clinical medical judgment.